

Termite construction

The Eastgate Centre in Harare, Zimbabwe which is inspired from termite mounds, has no air conditioning yet has regulated temperatures



Vertical Limit

A single building 3,281 ft tall, 1,312 ft and a total floor area of 8 square kilometers. The concept building is to have a capacity of about one lac workers! Can it get any bigger?

Feature

of concrete is used for the building, by comparison a typical water tank in a household is of 1 cubic metre capacity. Constructing this massive building took a total of 22 million man-hours.

The climate of the construction site provided added challenges due to the heat in the daytime with temperatures exceeding 50 degree Celsius. Therefore, the concrete was poured during the night (the air is much cooler and the humidity is higher, which is essential for curing the mixture evenly) mixed with ice and was pumped to a staggering height of 601 metres creating a world record of its own. Previous record for concrete pumping was held of the *Riva del Garda Hydroelectric Power Plant* in Italy in 1994, when concrete was pumped to a height of 532 metres. At the peak of its construction, the *Burj Khalifa* was adding a story to the structure every three days and hence it was absolutely essential for the concrete mix to not form any cracks and cause any delays.

Constructing structures so far above the ground faces serious wind problem. To give an example, the wind velocity at the top of the *Burj Khalifa* can be thrice the velocity at the base. Given the climate of Dubai, wind speeds at the base itself can exceed 50 kilometres per hour. The Chicago-based architectural firm Skidmore, Owings & Merrill, who were responsible for the design of *Burj Khalifa* were aware of this challenge and to counter it they had contracted a Canadian-based wind engineering firm RWDI, who was responsible for checking if the various designs of the building made by the architects were

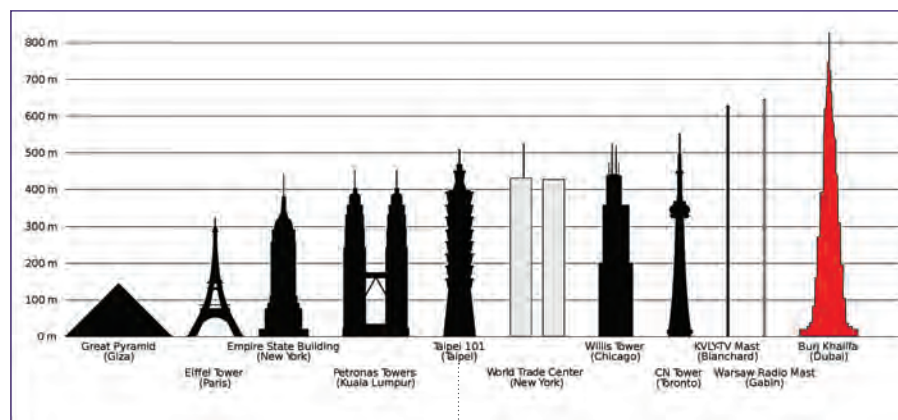
aerodynamically efficient. According to Wayne Boulton, GM (Wind Energy) at RWDI, "We constructed a scale model and put it in a wind tunnel. In the wind tunnel, we are able to test a number of different wind speeds and directions. We can test the pressure you would get on the surface of the building under normal conditions and also under more extreme events."

Safety

With more than 10,000 people living in a confined space of a modern day skyscraper safety becomes an important

sounding device is actually a 730 ton ball suspended by eight solid steel cables. This device acts like a giant pendulum and counters the motion of the building due to seismic and atmospheric forces. It was designed and constructed by firm called Motioneering. It can move to a maximum of five feet in any direction and is surrounded by a protective bumper ring to prevent it from swaying too far.

Another example of the installation of a Tuned Mass Damper is an oil based hydraulic system in the Citicorp Center in New York. The hydraulic system adjusts



A graphic representation of Skyscraper heights along the ages

issue. There are two aspects for it – one is protecting from man-made mishaps such as fire and electricity, which is relatively easier to tackle. The other is protecting a structure more than 500 metre above the ground from forces of nature such as typhoons and earthquakes. We shall look at two counter systems to get a feel of structural safety measures in skyscrapers.

The Taipei 101 Tower sits merely 660 feet from a major fault line in Taiwan, or in simple terms is very prone to earthquake damage. To counter this, the construction engineer first wrapped the entire building in a massive steel cage of immensely welded steel columns that is designed to flex and bend to absorb seismic energy. The engineers proclaim that this can withstand the impact of the most powerful earthquake in 2,500 years.

A second counter measure is the inclusion of a Tuned Mass Damper (TMD), which also prevents damage from lateral forces such as strong wind. This innocent

the position of the a 400-ton concrete weight block located on one of the top floors and in effect adjusting the entire weight of the building from one side to another. A powerful array of computers and sensors monitor the wind forces around the building and generate the stimulus for the movement of the block.

Burj Khalifa also has the provision of pressurised, fire-proof, structurally reinforced and critically air-conditioned refuge chambers located after every 25 floors, so in case of any disaster, natural or man-made disaster occupants needn't climb down more than 160 floors to find safety.

Another unique challenge is the skyscraper causing earthquakes in the surrounding region. Many had believed that the Taipei 101 tower had reactivated a dormant fault line. In order to counter this argument the designers and construction contractor of the tower had ensured that they remove equivalent amount of soil from the foundation as the



Extreme engineering solutions like Wind Tunnel simulations, generally reserved for race cars, are used in the construction of Modern Day Skyscrapers